EE 226: Semiconductor Devices

Lecture 4: Crystal Structure – Part 2
Energy Band Formation – Part 1

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Course Coverage

1. What is current? $I = \frac{dQ}{dt}$ Module 1
    1. $\boldsymbol{Q \Rightarrow electrons, holes \Rightarrow How\ many\ are\ there?}$
    2. How do they flow in a given device?
    3. Any other dependencies

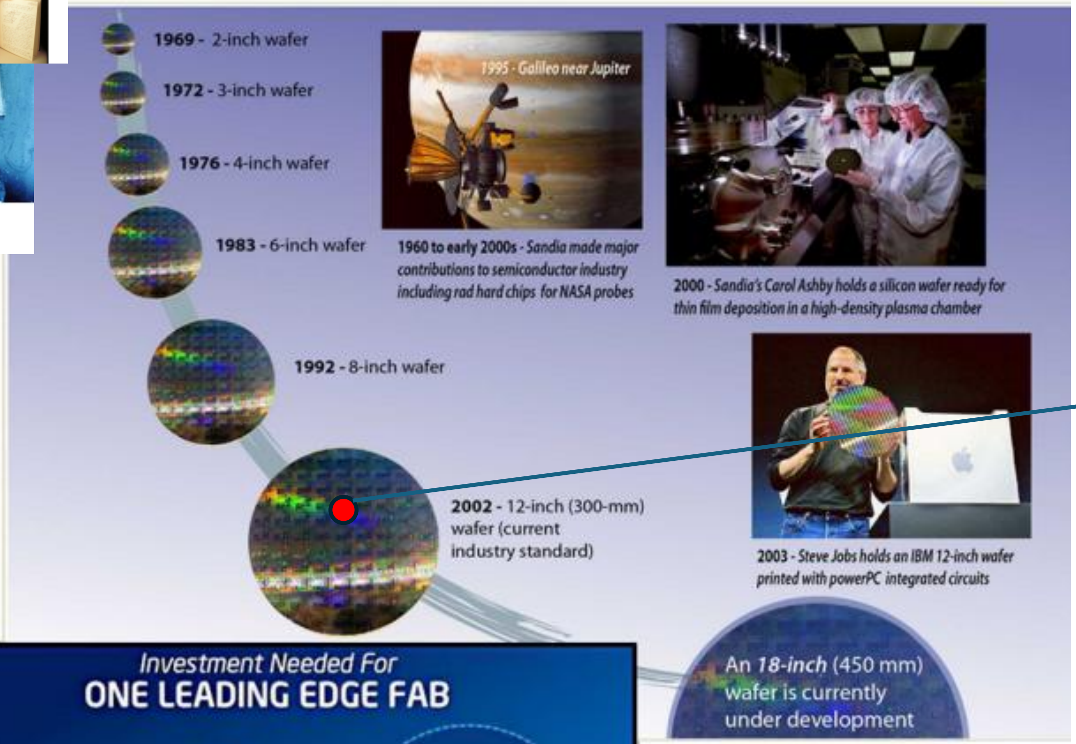
2. How does a diode work? Module 2
    1. All the questions raised in previous slide.

3. Why do we need a 3 (or 4)-terminal device & how does it work? Module 3, Module 4

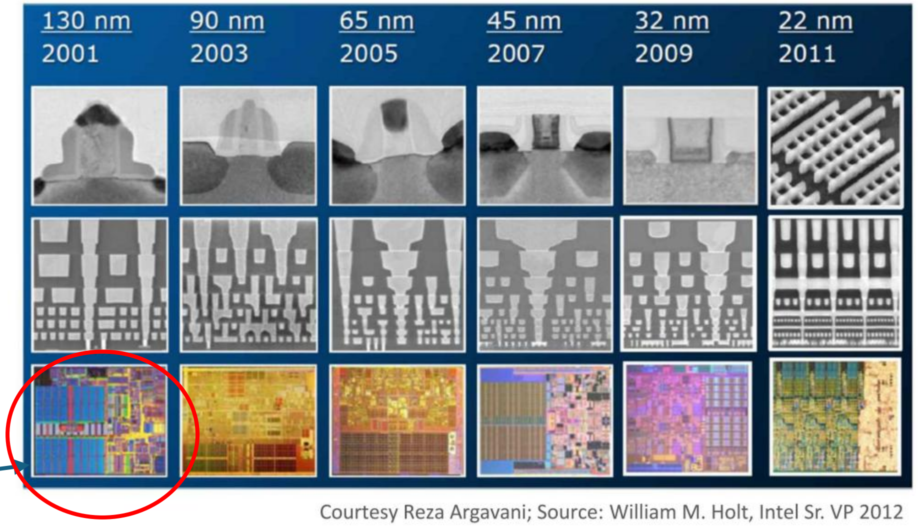
4. Need and operation of memory devices? Module 5

Recap- Computing – Transistors - MOSFETs

Jack Kilby
Texas Instruments

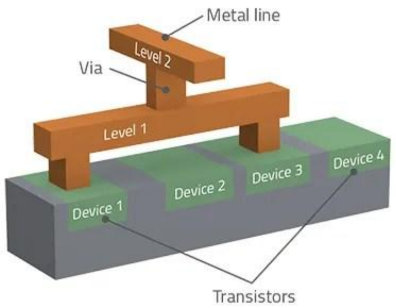


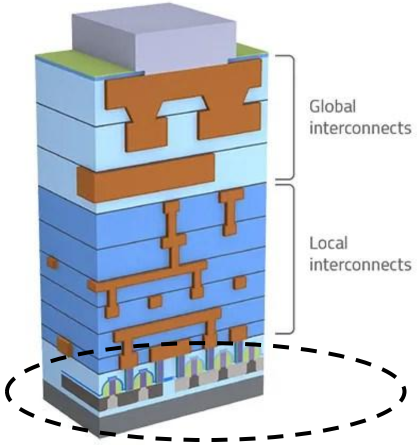
D. Nenni, 450mm Wafers are Coming!, SemiWiki.com, Aug 14 2013



Courtesy Reza Argavani; Source: William M. Holt, Intel Sr. VP 2012

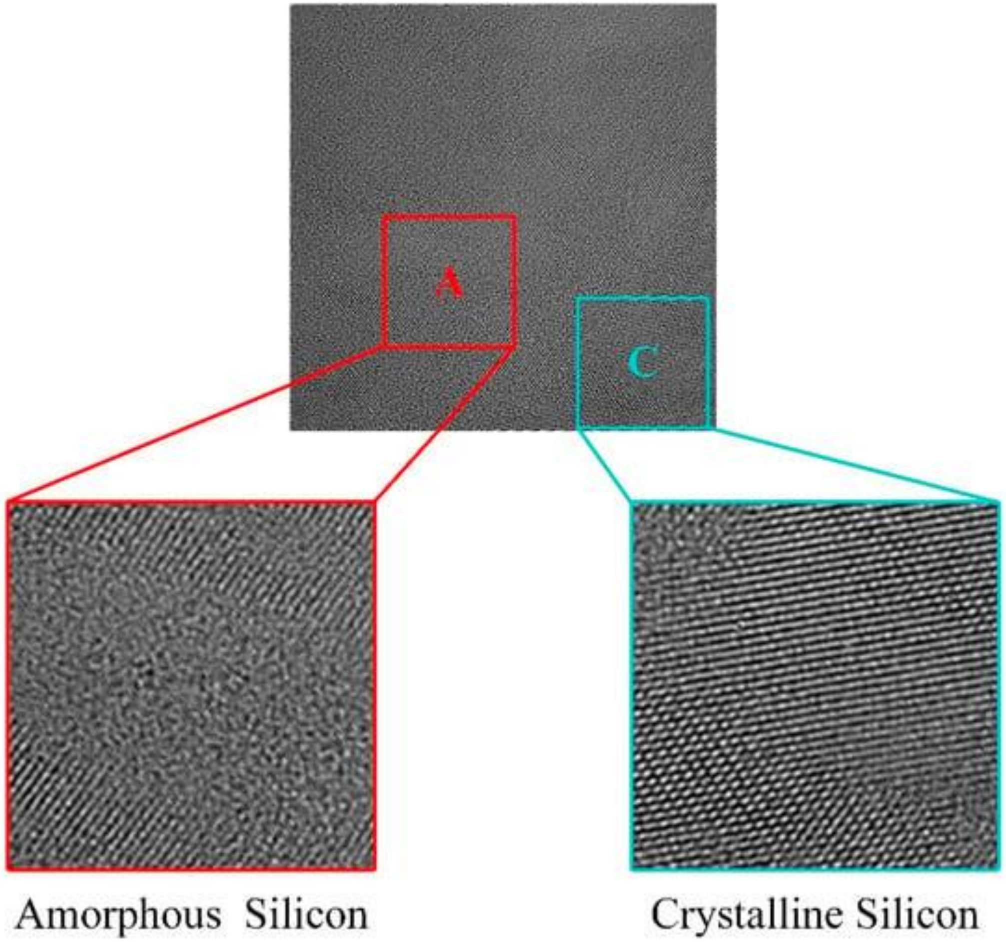
Cross sectional view



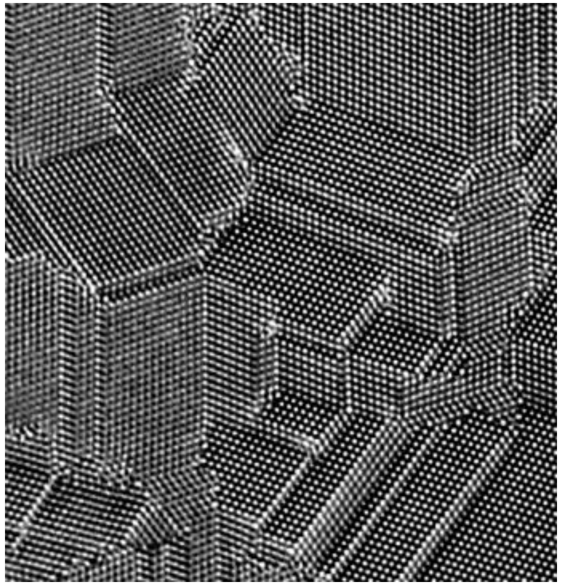


Transistors

Recap: Types of Solid - Examples



Polycrystalline

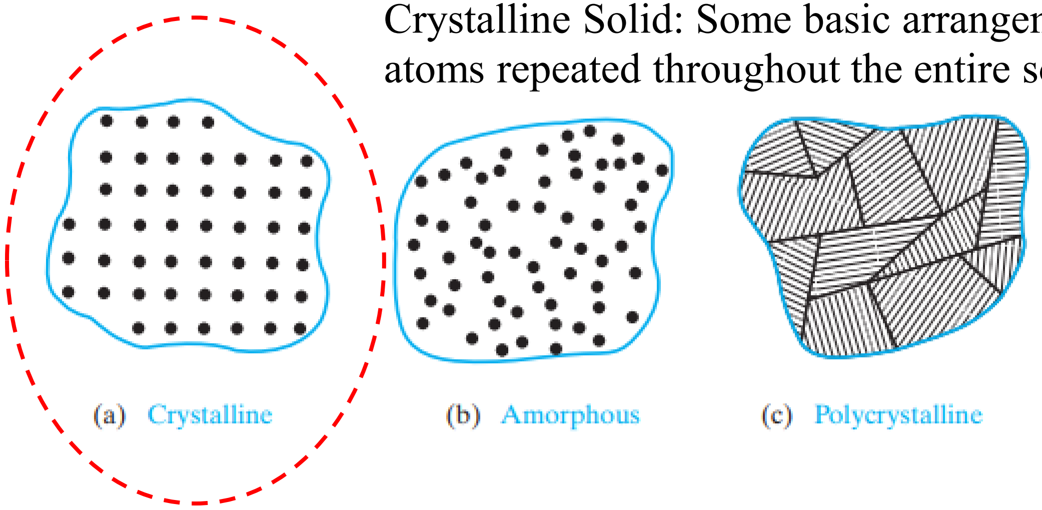


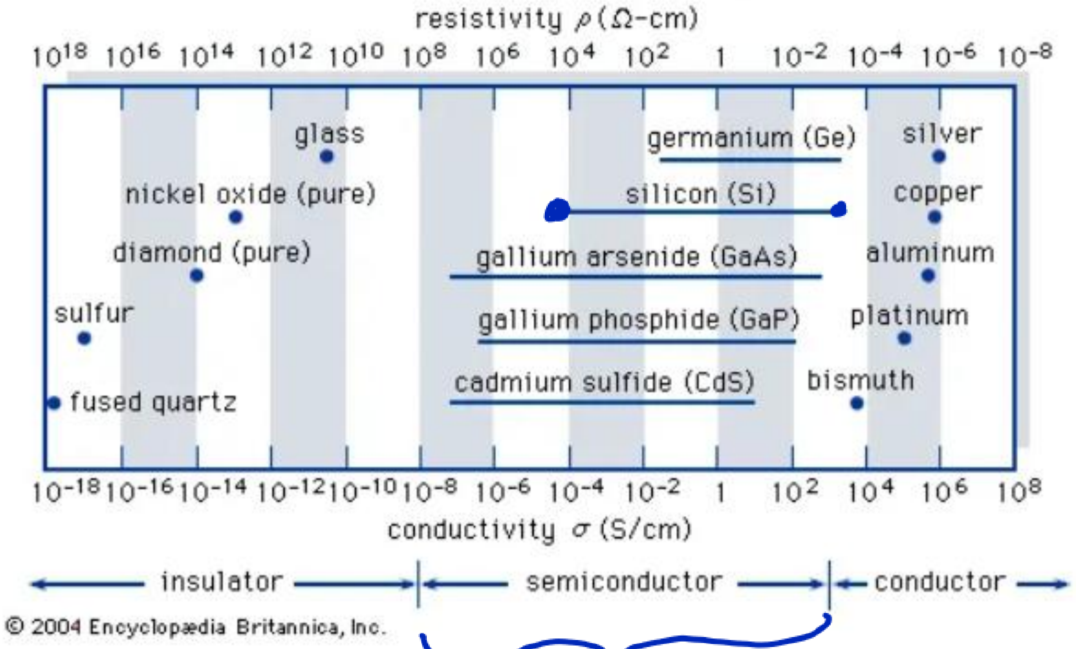
Recap: Why "Semiconductors"?

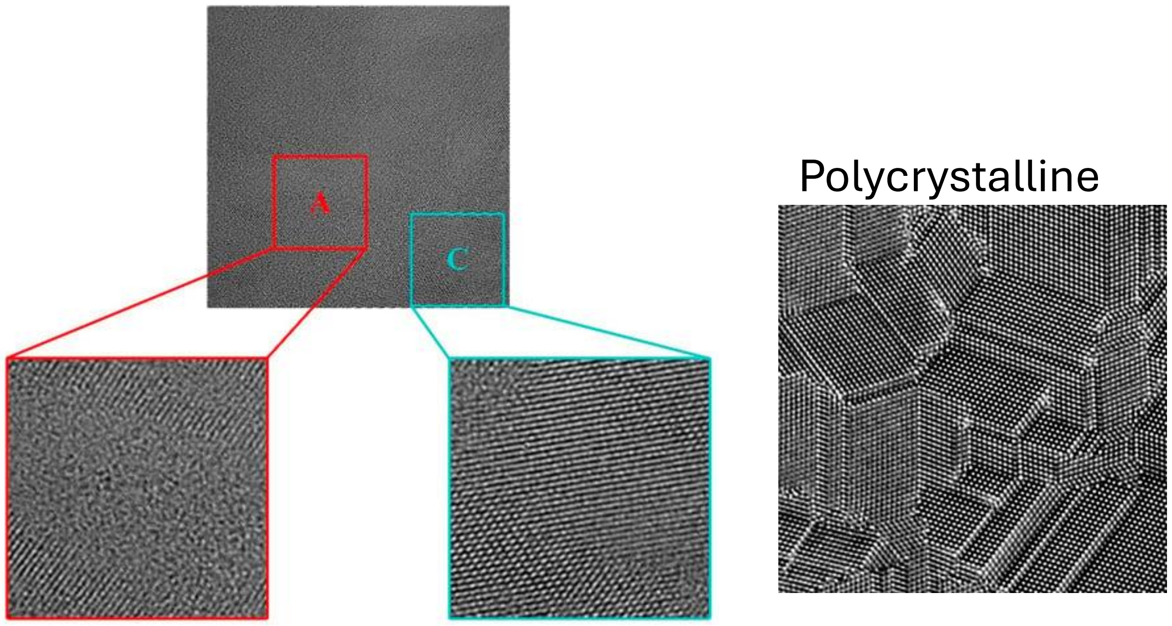
- Conductors – e.g Metals
- Insulators – e.g. SiO2 - Sand
- Semiconductors
  - ➢ Conductivity between conductors and insulators
  - ➢ Generally crystalline in structure

    In recent years, non-crystalline semiconductors have become commercially very important

Types of solid:
Crystalline Solid: Some basic arrangement of atoms repeated throughout the entire solid







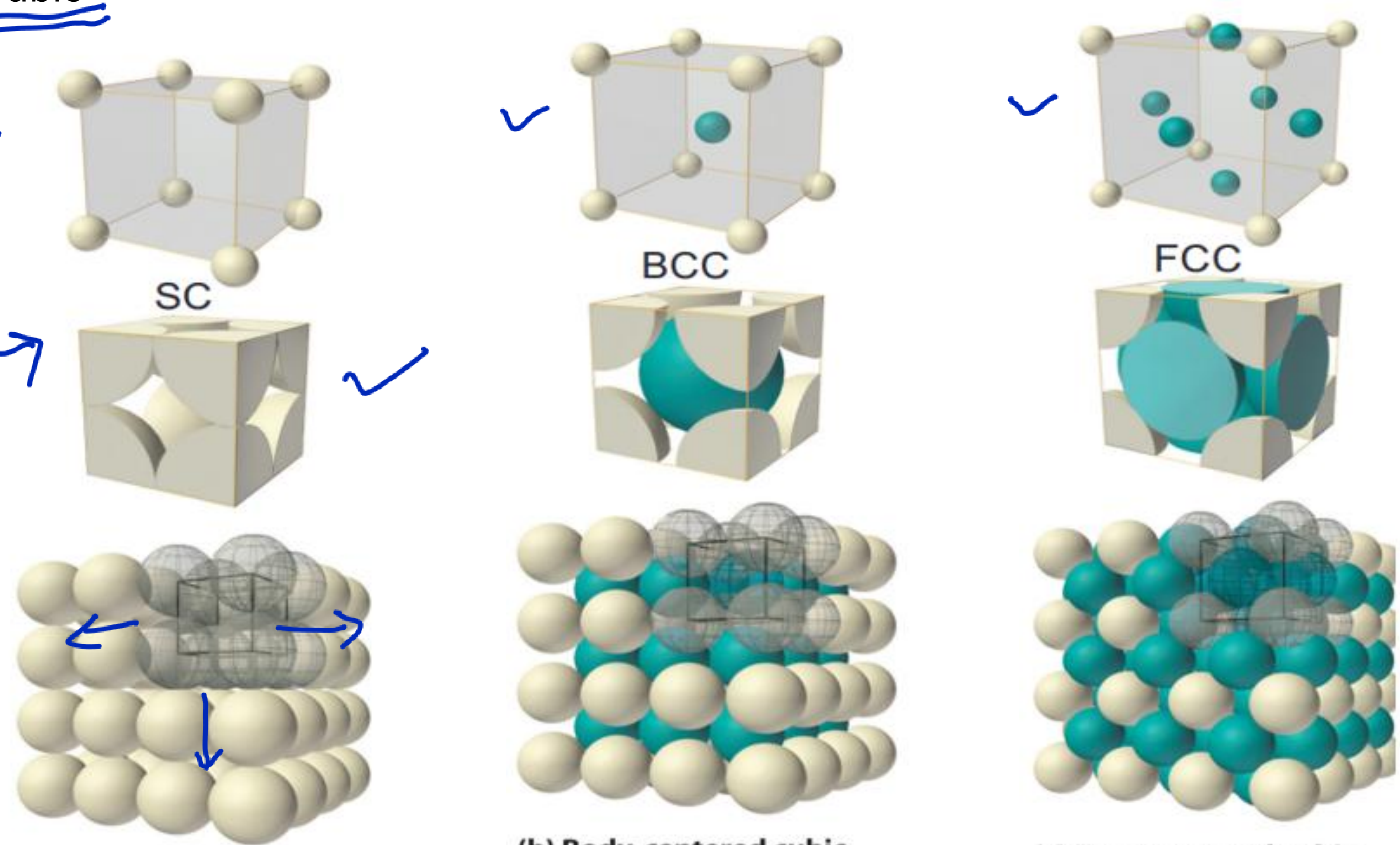
Amorphous Silicon   Crystalline Silicon

https://www.mdpi.com/2072-666X/13/5/743

Recap: Crystal Structure

lattice + Basis   Bravais

Cubic



(a) Simple cubic

(b) Body-centered cubic

(c) Face-centered cubic

- **Crystal Surfaces**
  - Planes
  - Directions
- **Packing in crystal**

Crystal Surfaces

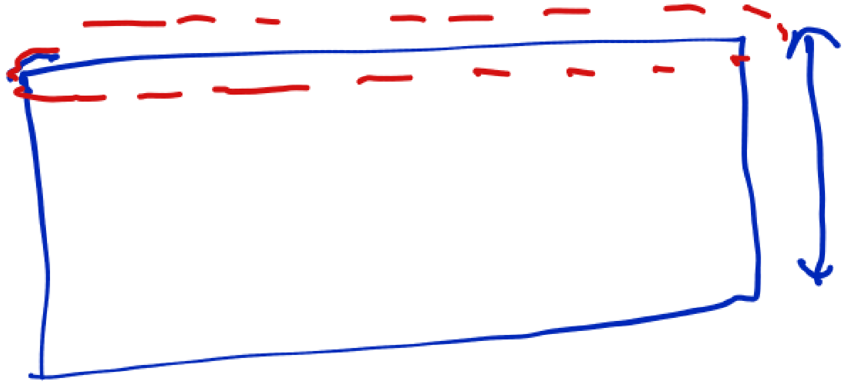
- Planes
- Direction

What information do we get from crystal surfaces?

- Atoms sit at specific positions and surfaces
- Surface of a wafer (material) is carefully oriented to lie along a specific plane
- Precise surface orientation(direction) affects characteristics of devices

#, movement.

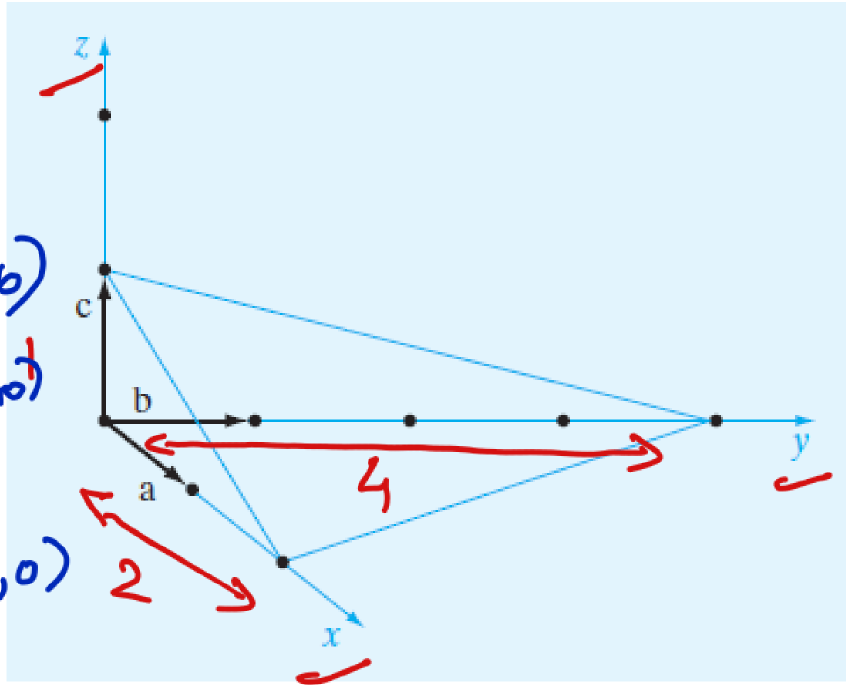
Miller indices: Naming convention of surfaces i.e. planes and direction



Crystal Surfaces: Miller Indices

- Planes

1. Find the intercepts of the plane with the crystal axes   2a, 4b, 1c
2. Express these intercepts as integral multiples of the basis vectors
   2, 4, 1   (2, 4, ∞)
3. Take the reciprocals of the three integers   1/2, 1/4, 1   (1/2, 1/4, 0)
4. Reduce to the smallest set of integers *h, k,* and *l*   4x   2, 1, 4   (2, 1, 0)
5. Label the plane (*hkl*).   (214)

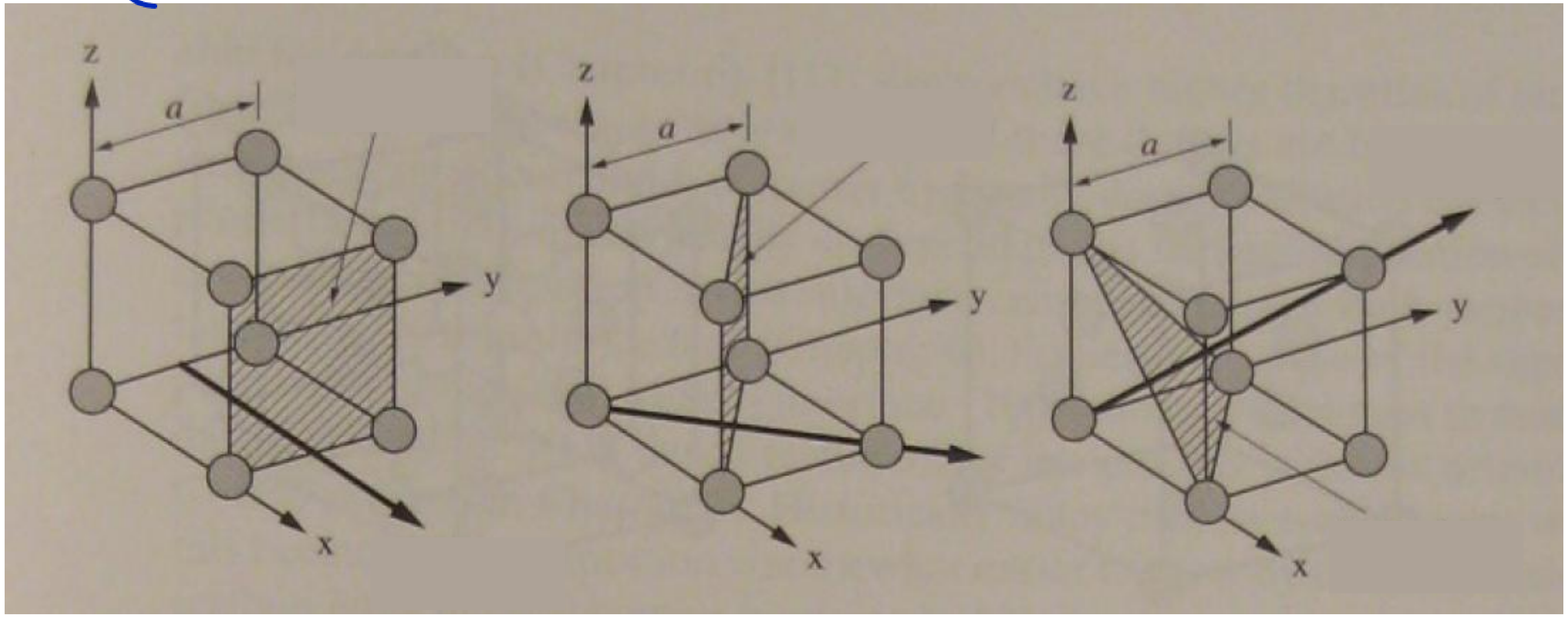


-ve intercept: ($\bar{2}$14)

Crystal Surfaces: Miller Indices

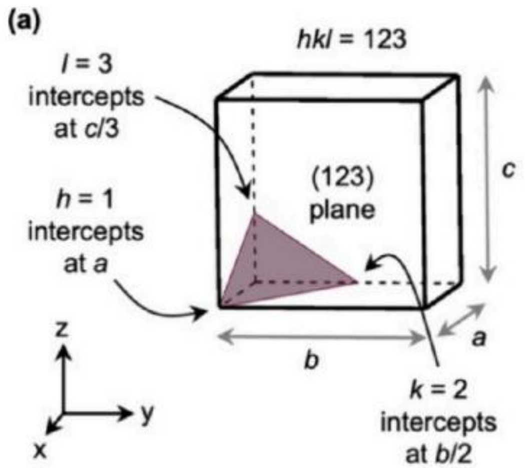
Some examples:

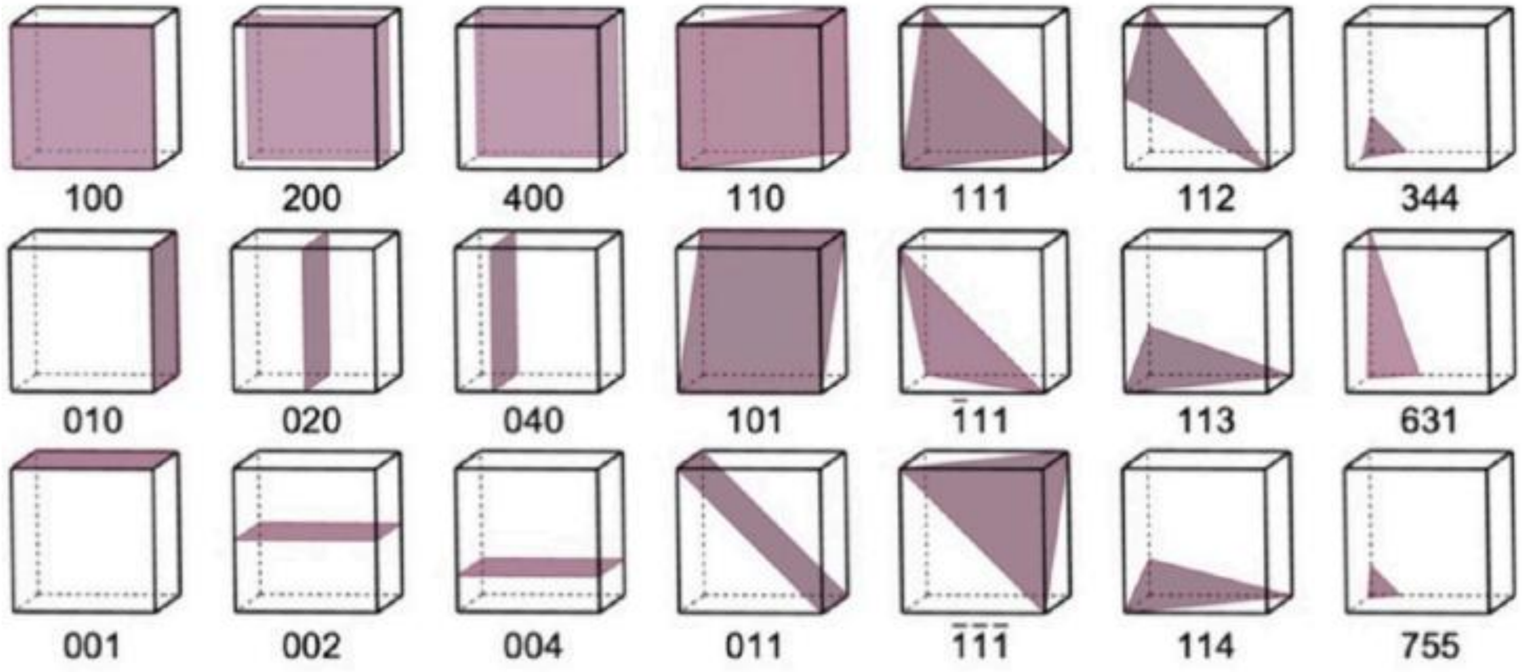
(100) (110) (111)



Crystal Surfaces: Miller Indices

More examples:



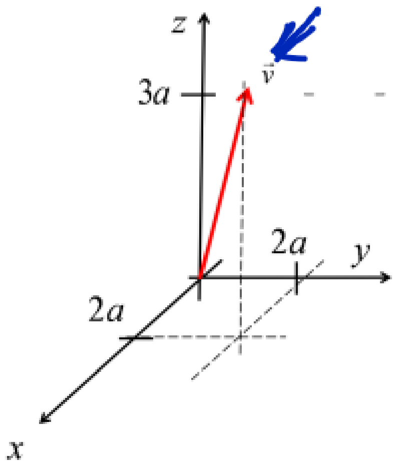


Tutorial!

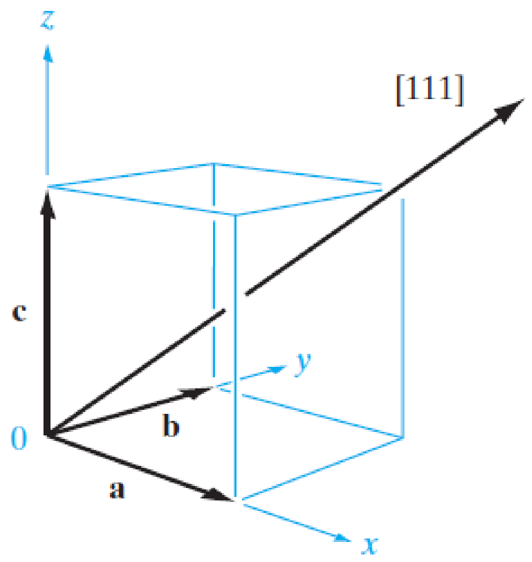
Crystal Surfaces: Miller Indices

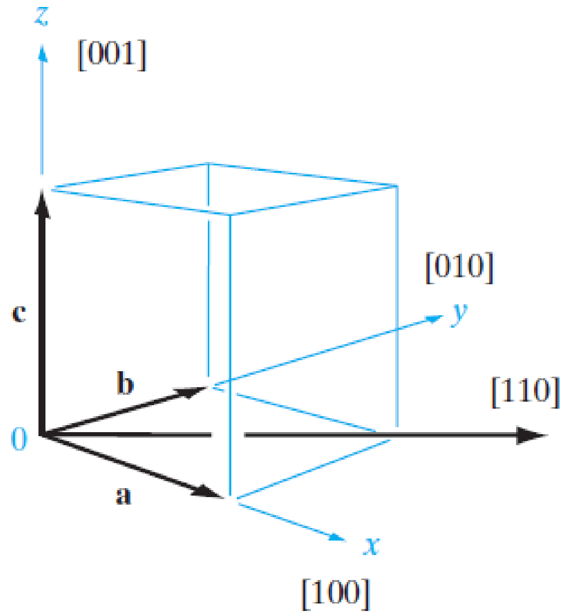
- Direction

A direction in a lattice is expressed as a set of three integers with the same relationship as the components of a vector in that direction



1. Equation of vector
2. Describe the components e.g. h,k,l   2, 2, 3
3. Represent in [hkl]   [223]



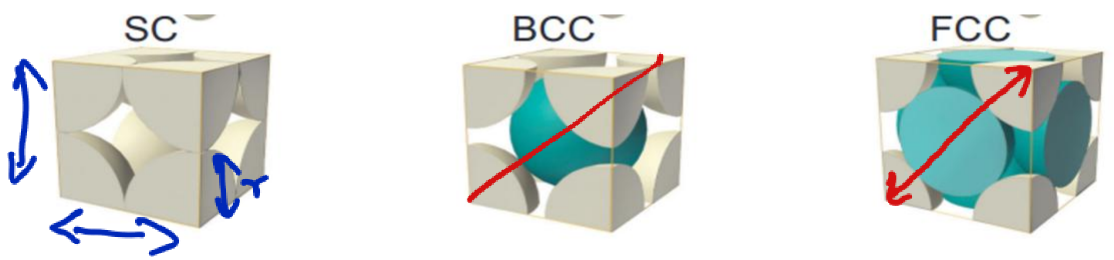


Packing in Crystals

"cm"

$\left(\#/cm^3\right)$ Volume

$\left(\#/cm^2\right)$ surface



a
lattice constant

1) Number of atoms

2) Atomic Packing fraction (APF)

$$APF = \frac{N \times V_{atom} \left(\frac{4}{3}\pi r^3\right)}{V_{unit\ cell} \left(a^3\right)}$$

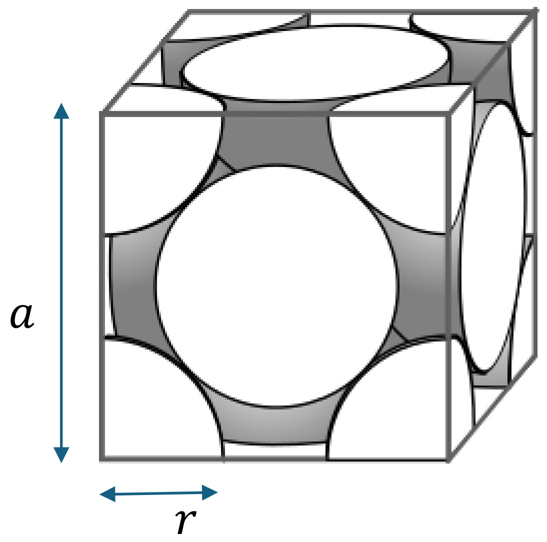
SC	BCC	FCC
$a = 2r$	$a =$	$a =$
$N = \frac{1}{8} \times 8 = 1$	$N = \frac{1}{8} \times 8 + 1 = 2$	$N = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$
$\sim 52\%$	$\sim 68\%$	$\sim 74\%$

1) N

2) Rel^n betn r & a

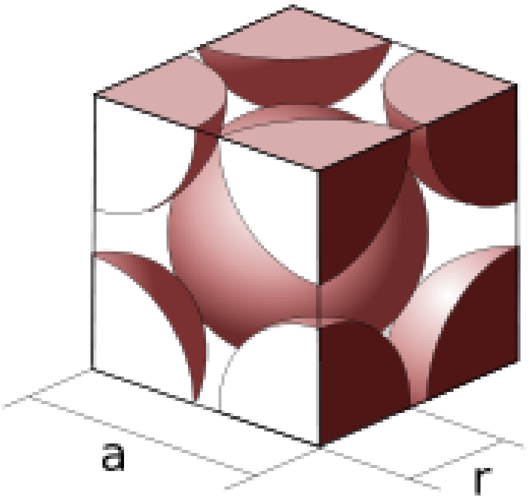
Packing in Crystals

Face-centered cubic (fcc)



$$a = 2r\sqrt{2}$$

Body-centered cubic (fcc)



$$a = \frac{4r}{\sqrt{3}}$$

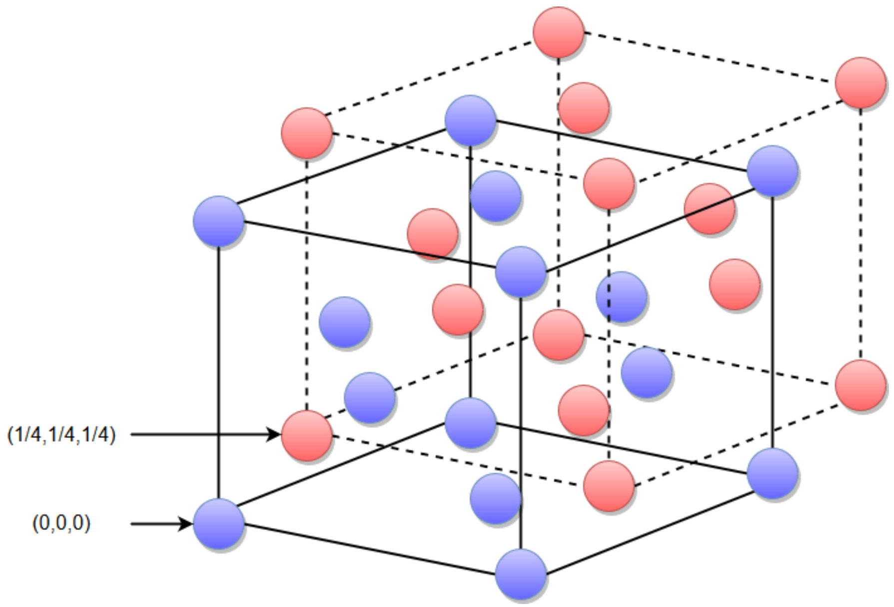
1. APF?
2. What about Volume Density?
3. Areal Density in (100) plane?

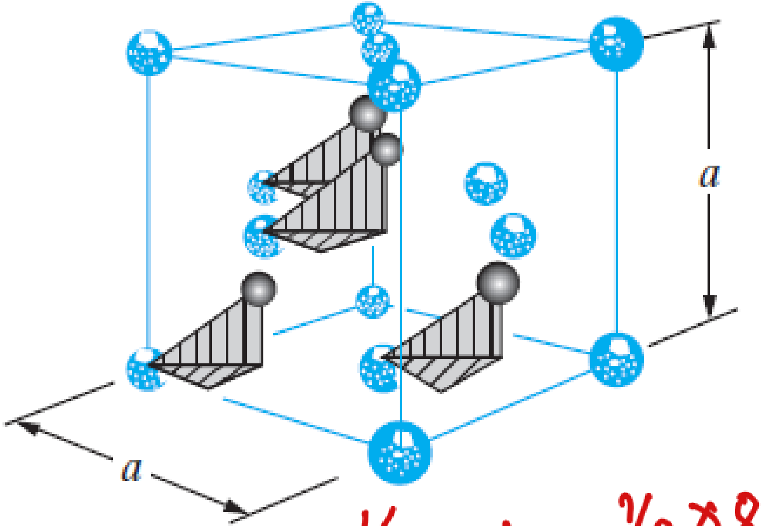
Tutorial Questions!

https://en.wikipedia.org/wiki/Atomic_packing_factor

Diamond Crystal Structure – Si, Ge

The diamond structure can be thought of as an fcc lattice with an extra atom placed at **1**/4 + **1**/4 + **1**/4 from each of the fcc atoms.





Can this be called Bravais lattice?

lattice + Basis

FCC (with 2 basis atoms)

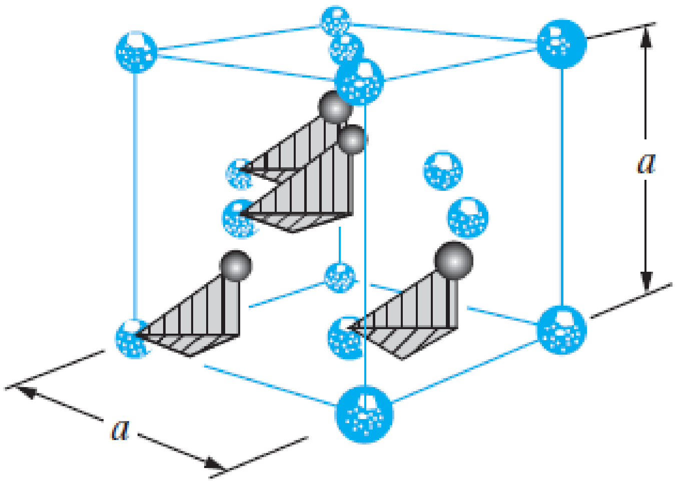
$N = \frac{1}{8} \times 8 + \frac{1}{2} \times 6$

$+ 4$

$N = 8$

ans

Diamond Crystal Structure – Si, Ge



Given a = 5.43 A^{O}

1. What is the atomic packing fraction (APF) for the diamond crystal structure? APF

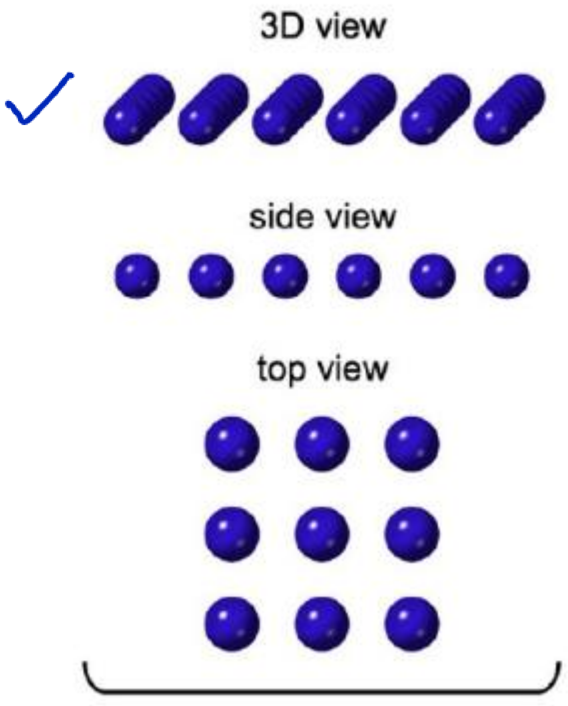
2. What is the volume density $(\frac{\text{number}}{cm^3})$ for the diamond crystal structure?

3. What is the areal density $(\frac{\text{number}}{cm^2})$ for the diamond crystal structure in (100) plane?

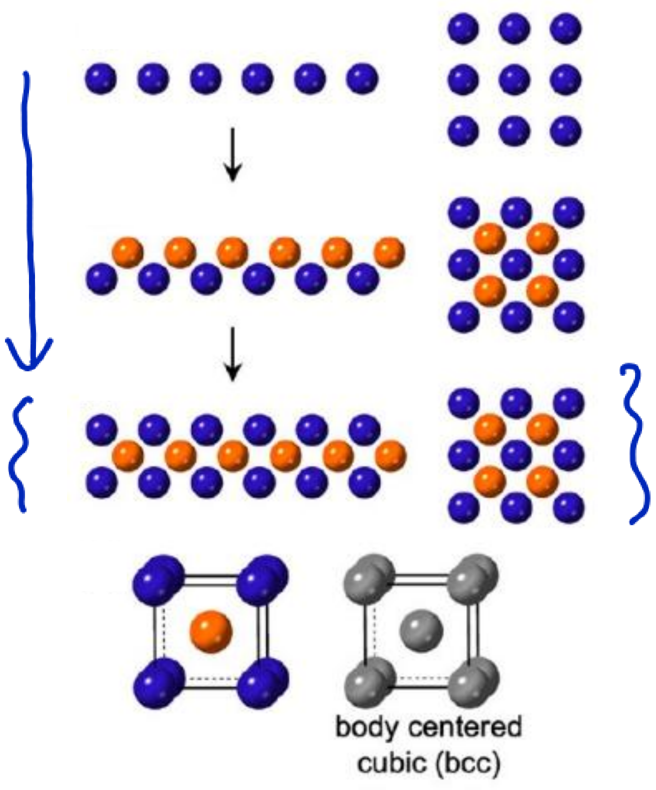
More Tutorial Questions !

Visualizing different crystals – 1D, 2D, 3D

Simple Cubic



Body Centered Cubic



pubs.acs.org/nanoau

Review

Tutorial on Describing, Classifying, and Visualizing Common Crystal Structures in Nanoscale Materials Systems

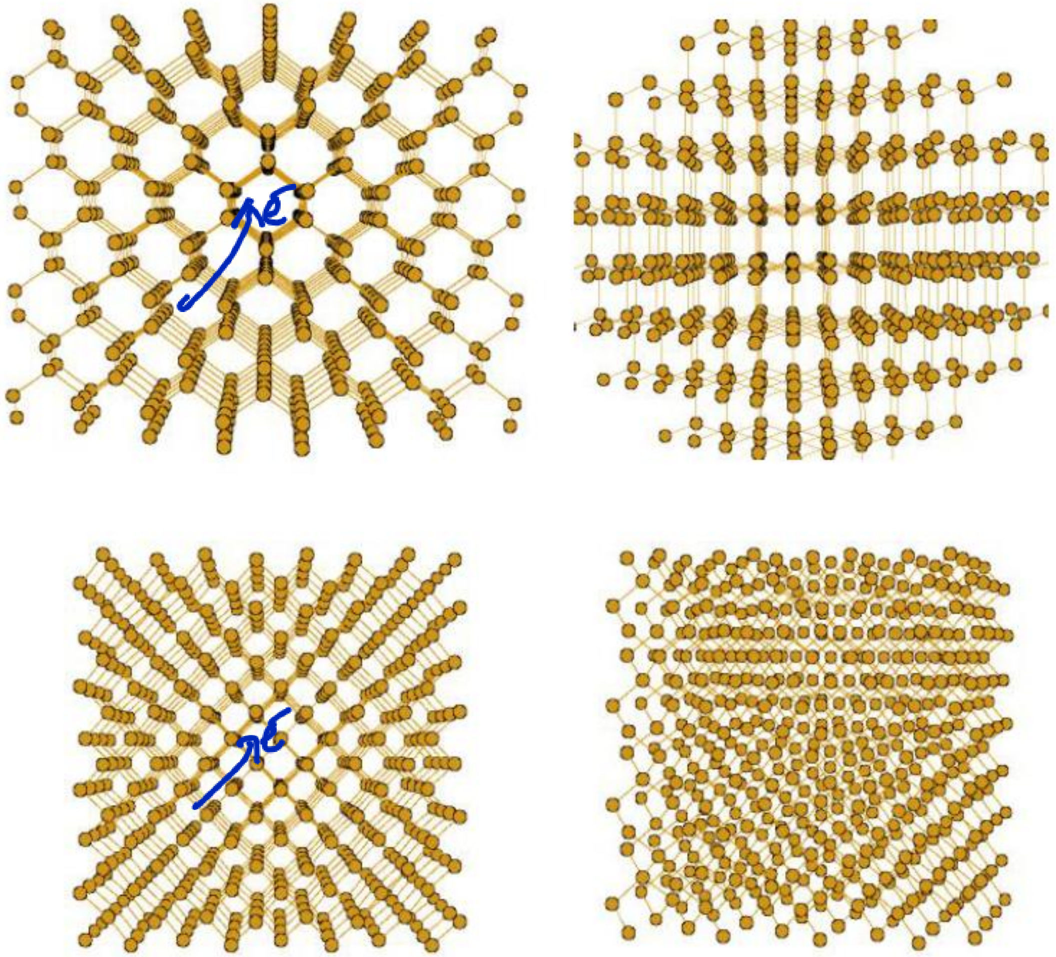
Katelyn J. Baumler and Raymond E. Schaak*

Cite This: ACS Nanosci. Au 2024, 4, 290–316

Read Online

https://pubs.acs.org/doi/pdf/10.1021/acsnanoscienceau.4c00010?ref=article_openPDF

Diamond Structure– viewing through different planes

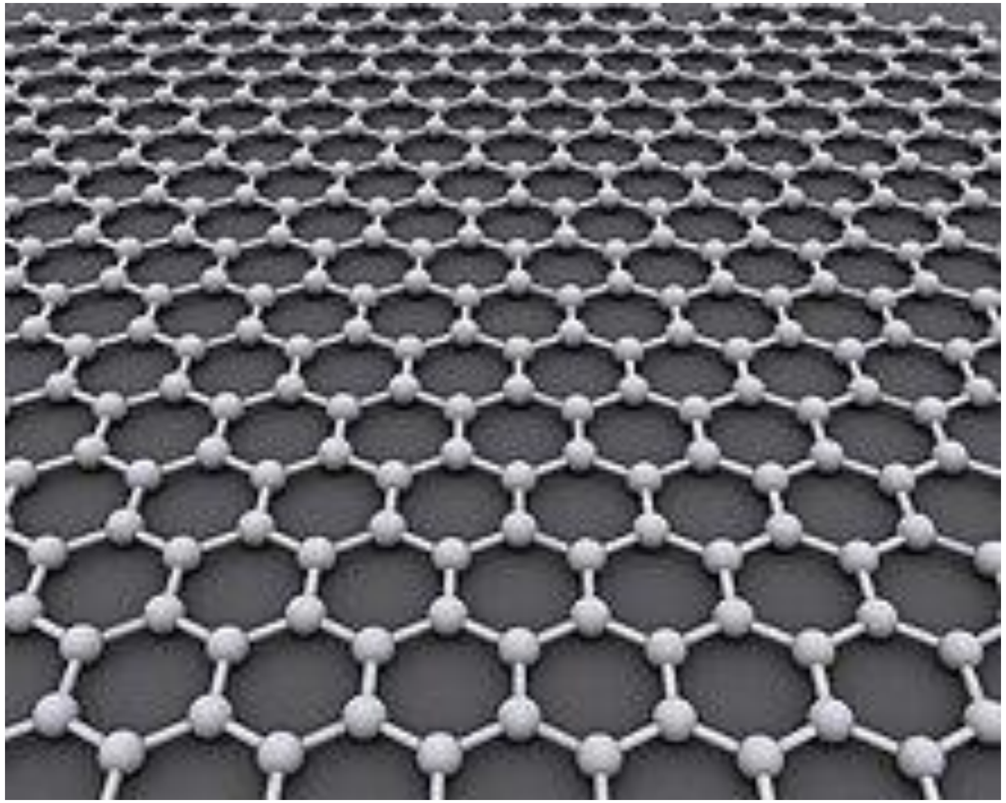


1) How does it affect
(#/cm²) ←

2) e⁻ movement

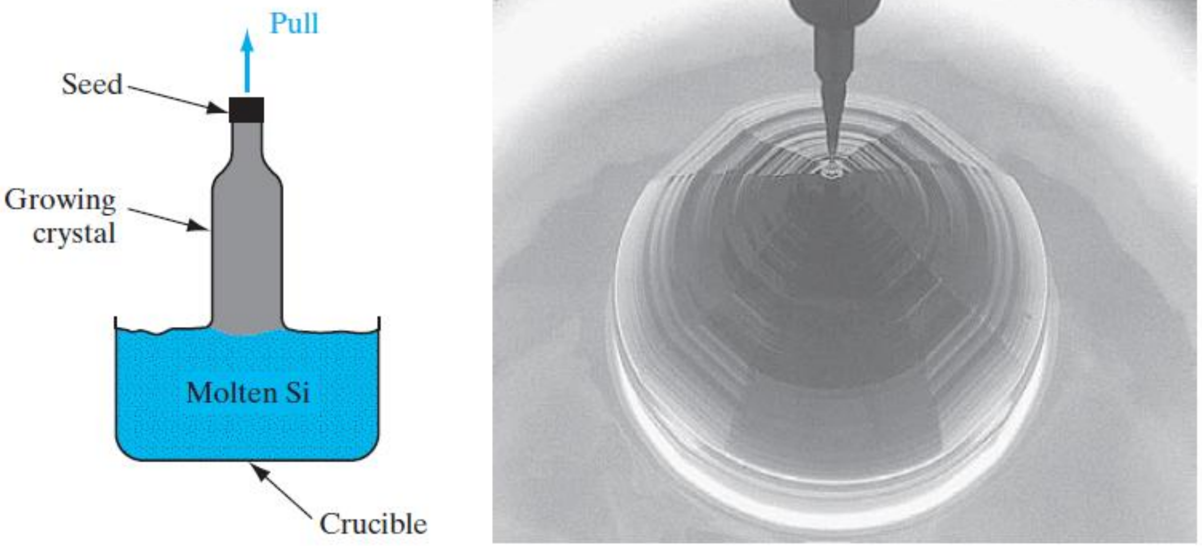
2D Crystal Lattice

Graphene is an atomic-scale honeycomb structure made of carbon atoms

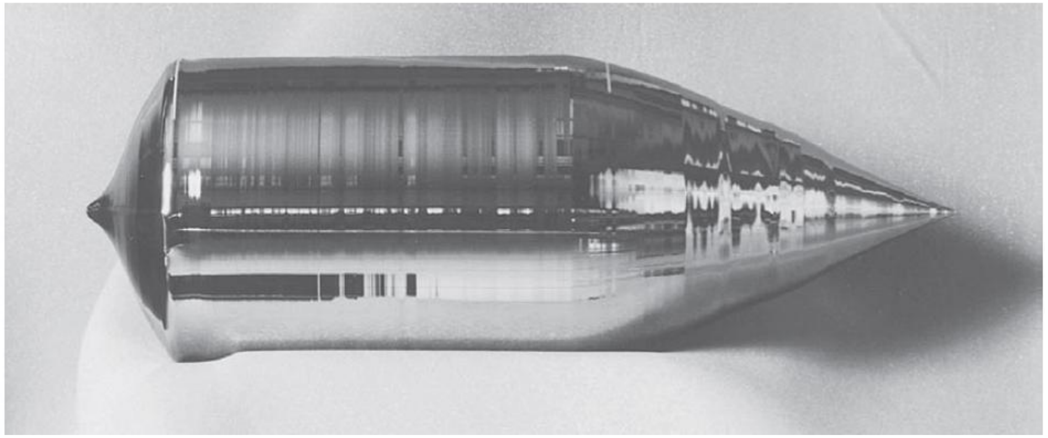


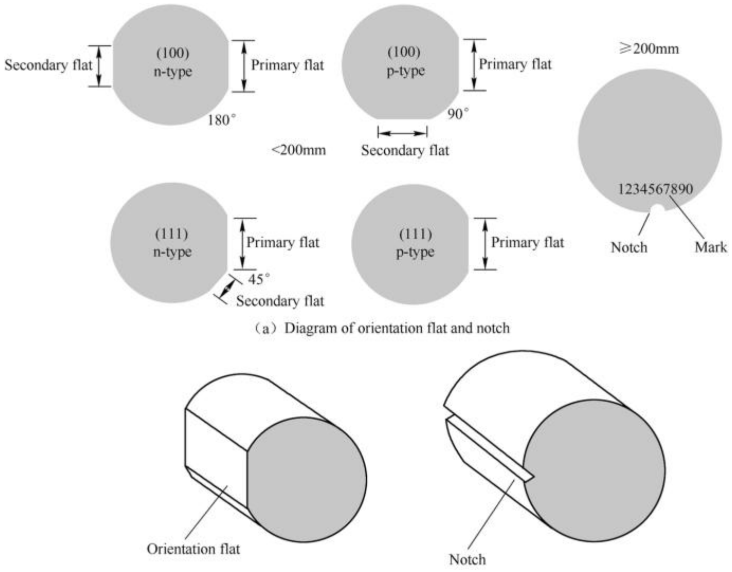
Andre Geim and Konstantin Novoselov, Nobel Prize in Physics 2010

Wafer fabrication process



(a) (b)





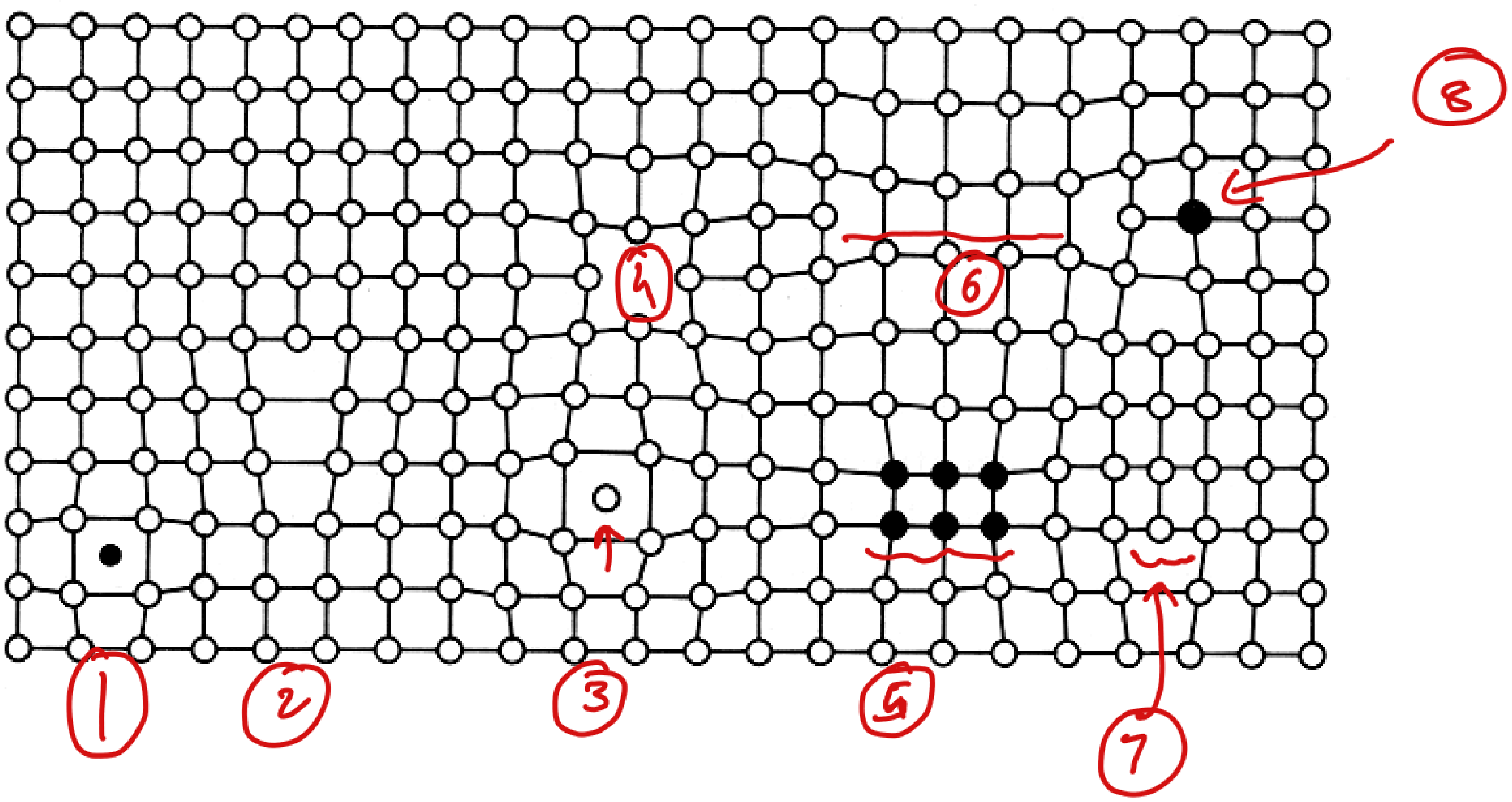
(a) Diagram of orientation flat and notch

(b) Axonometric diagram

Is the crystal perfect ?

Every crystal will have some defects inside

How many defects types can you identify?

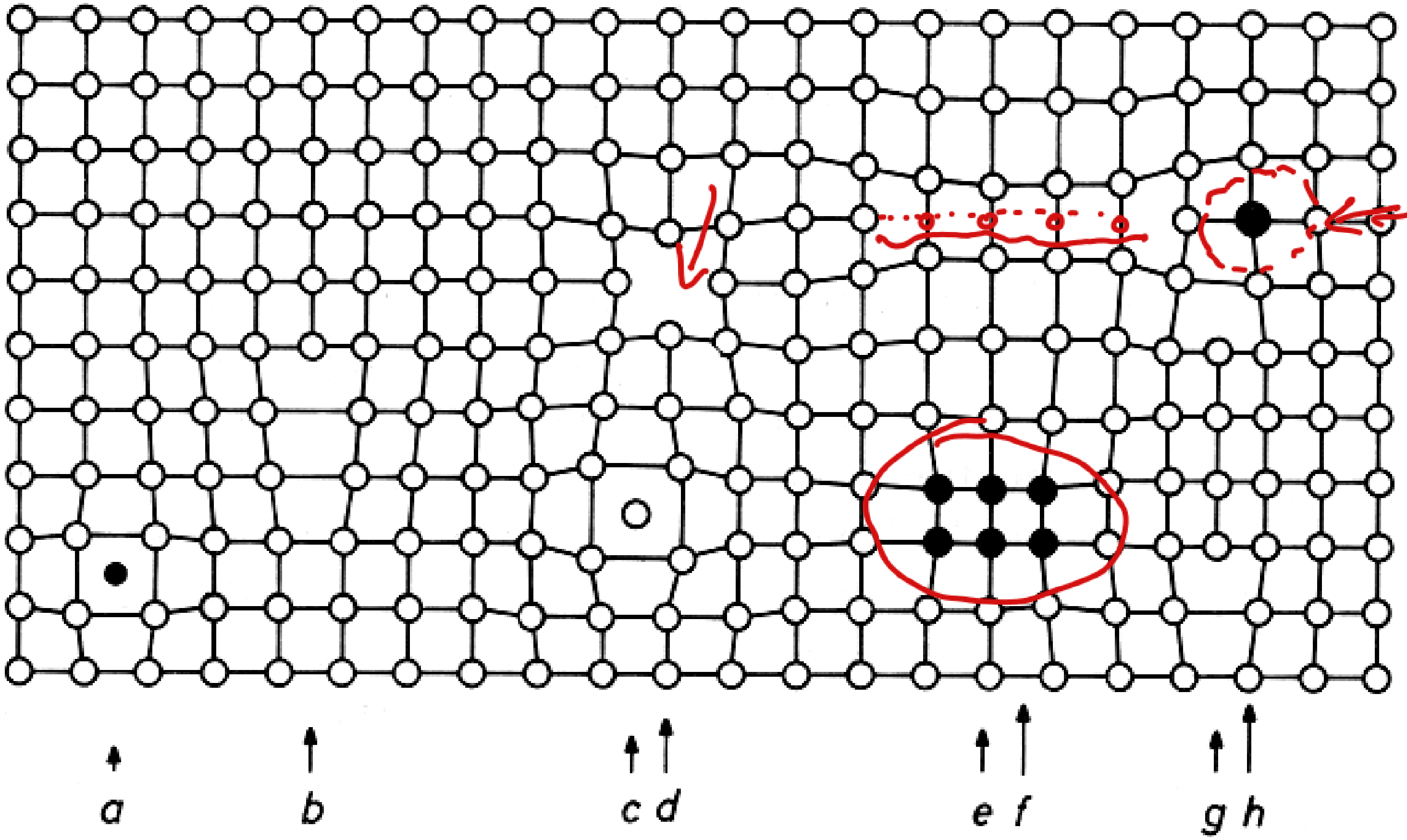


Is the crystal perfect ?

How many defects types can you identify?

1) Are defects useful?

→ Some defects are useful



a: interstitial impurity atom; *b*: edge dislocation; *c*: self interstitials; *d*: vacancy; *e*: precipitate of impurity atoms; *f*: vacancy-type dislocation loop; *g*: interstitial-type dislocation loop; *h*: substitutional impurity atom.

How pure is Silicon used today?

diamond : (#/cm³)

$\sim 10^{22}$ /cm³

10^{7}

- Metallurgical-grade silicon – 99% – alloys like ferrosilicon and aluminum-silicon alloys

- Solar-grade silicon – typically 99.9999%, or 6N – Solar Cells

- Electronic-grade silicon – 99.9999999% (9N or higher) – Semiconductor industry

10^{3} defects / cm^{3} 1 defect every 10^{9} : $\sim 10^{10}$

Find the number of defects ($/cm^3$) for all these grades of silicon?

Silicon – Finding Free Carrier Concentration

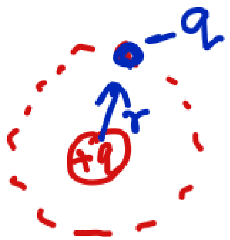
Si : $\sim 10^{22} / cm^3$

Si : 14 : $\sim 14 \times 10^{22} / cm^3$

Valence e^- : $\sim 4 \times 10^{22} / cm^3$

" How many e^- are free ? "

Hydrogen Atom – Continuous energies



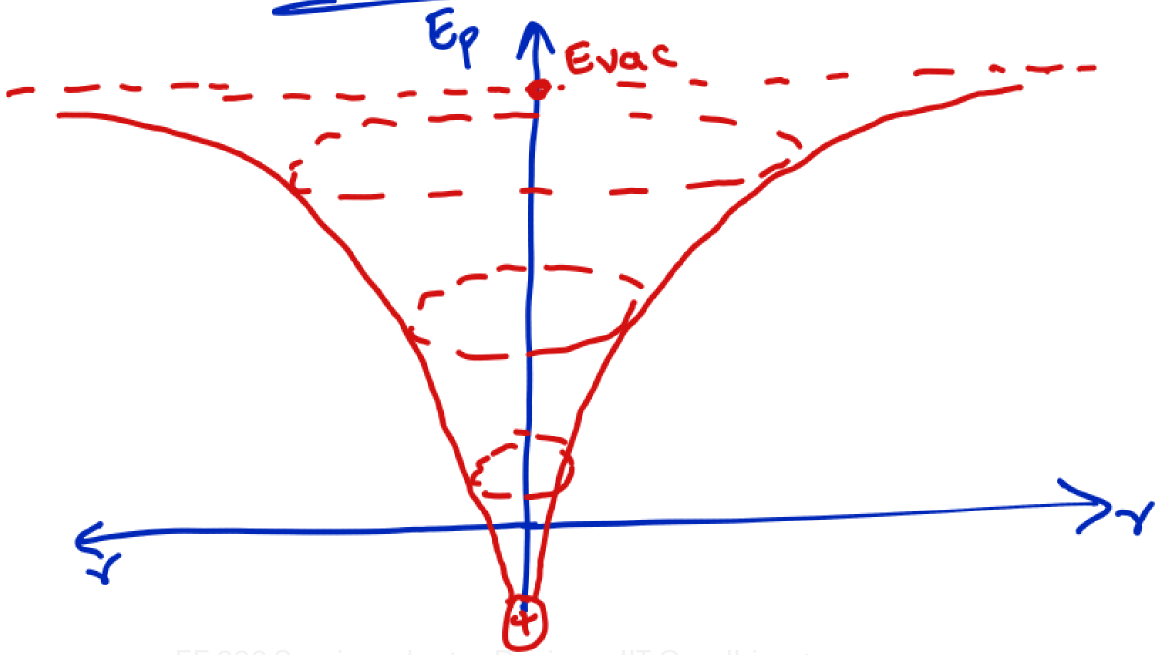
Coulomb's $F = -K \cdot \frac{q^2}{r^2}$

Potential Energy (E_p) : $F = -\frac{dE_p}{dr}$

$r: \quad r \to \infty$

$E_p: \quad E_p(r) \to E_{vac}$

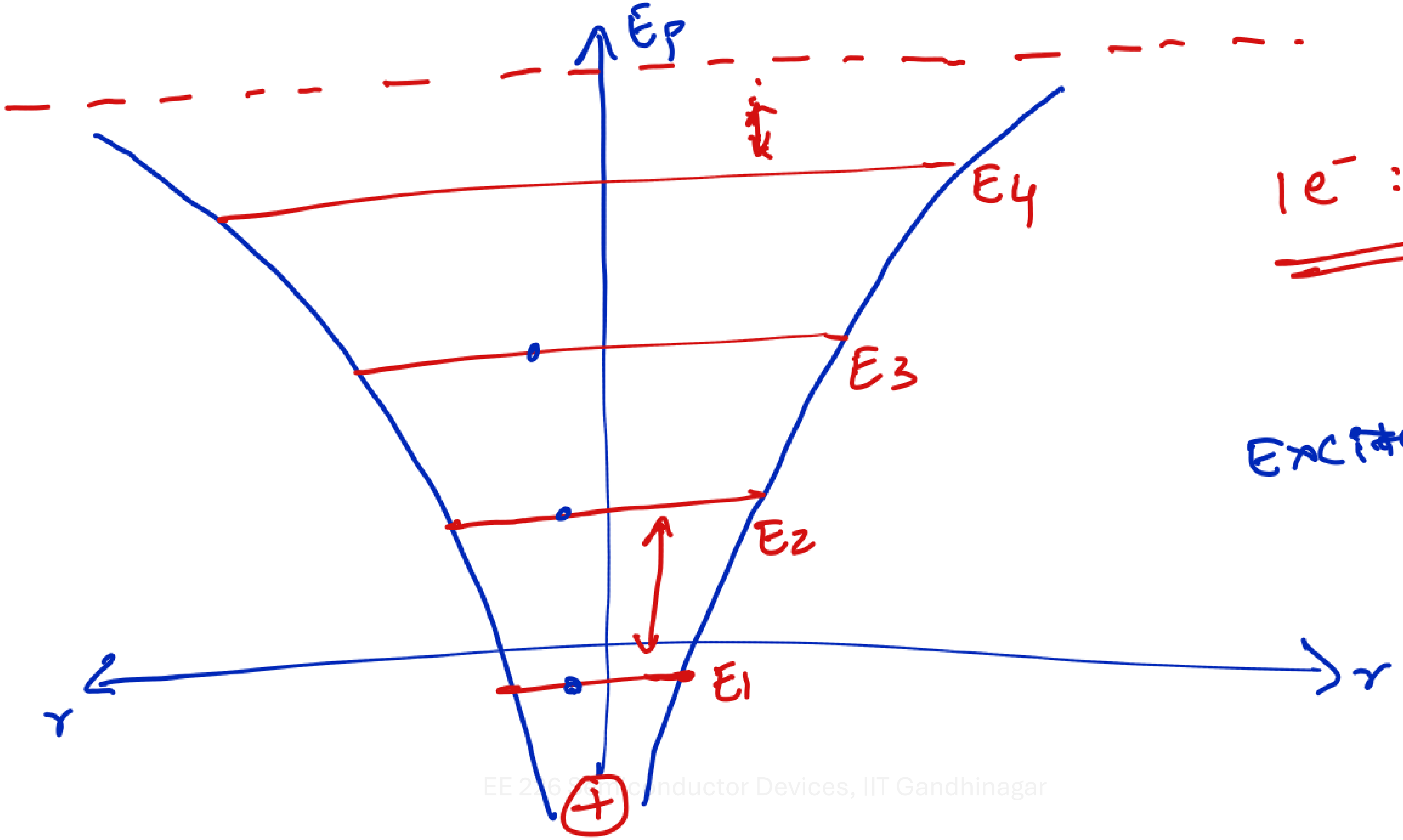
$$E_p = E_{vac} - K \cdot \frac{q^2}{r}$$



Hydrogen Atom – Quantized energy states

<u>Bohr's</u> : Energies are quantized defined by quantum number (n, k, l, s)

$$E_p = E_{vac} - k \cdot \frac{q^2}{n^2}$$



$1e^-$: N states

Excite : Energy

"T, V"

$1e^-$: N states

$2e^-$: $2N$ states

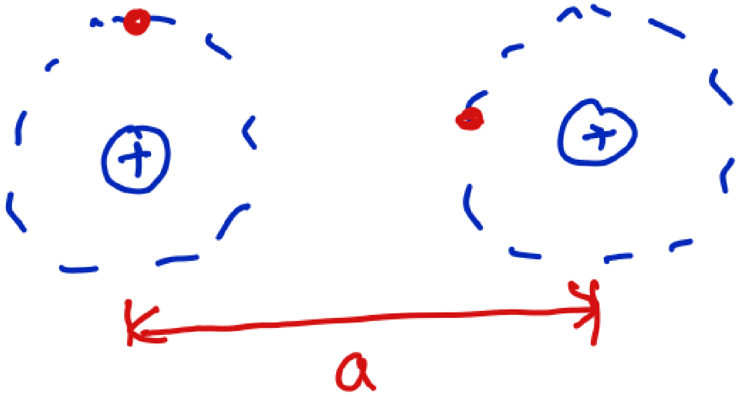
$4e^-$

⋮

$\sim 4 \times 10^{22}/cm^3$: ? ?

Two Hydrogen Atoms

How do hydrogen atoms form a covalent bond? (H_2)

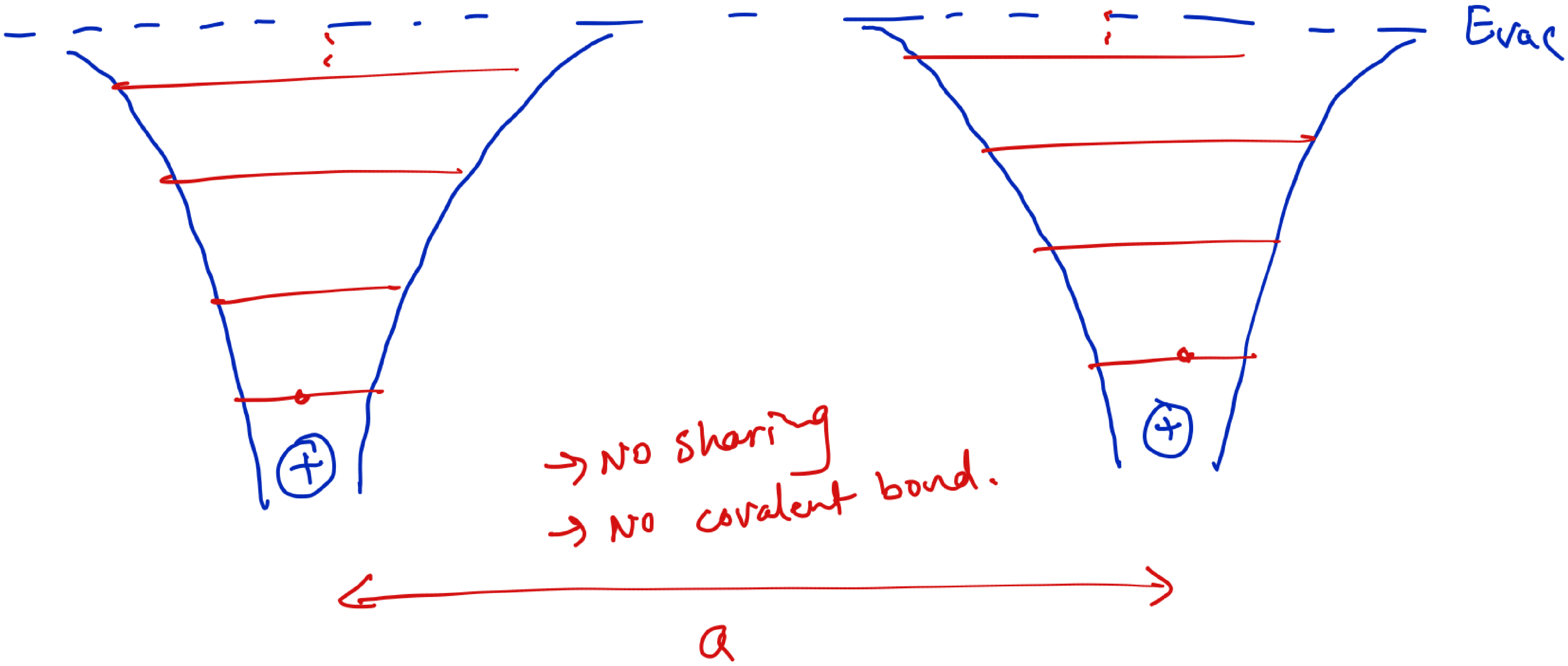


case 1: Non-interacting atoms (a large)

case 2: Interacting (a small)

case 3: Interacting (a very small)

case I: Non-interacting atoms

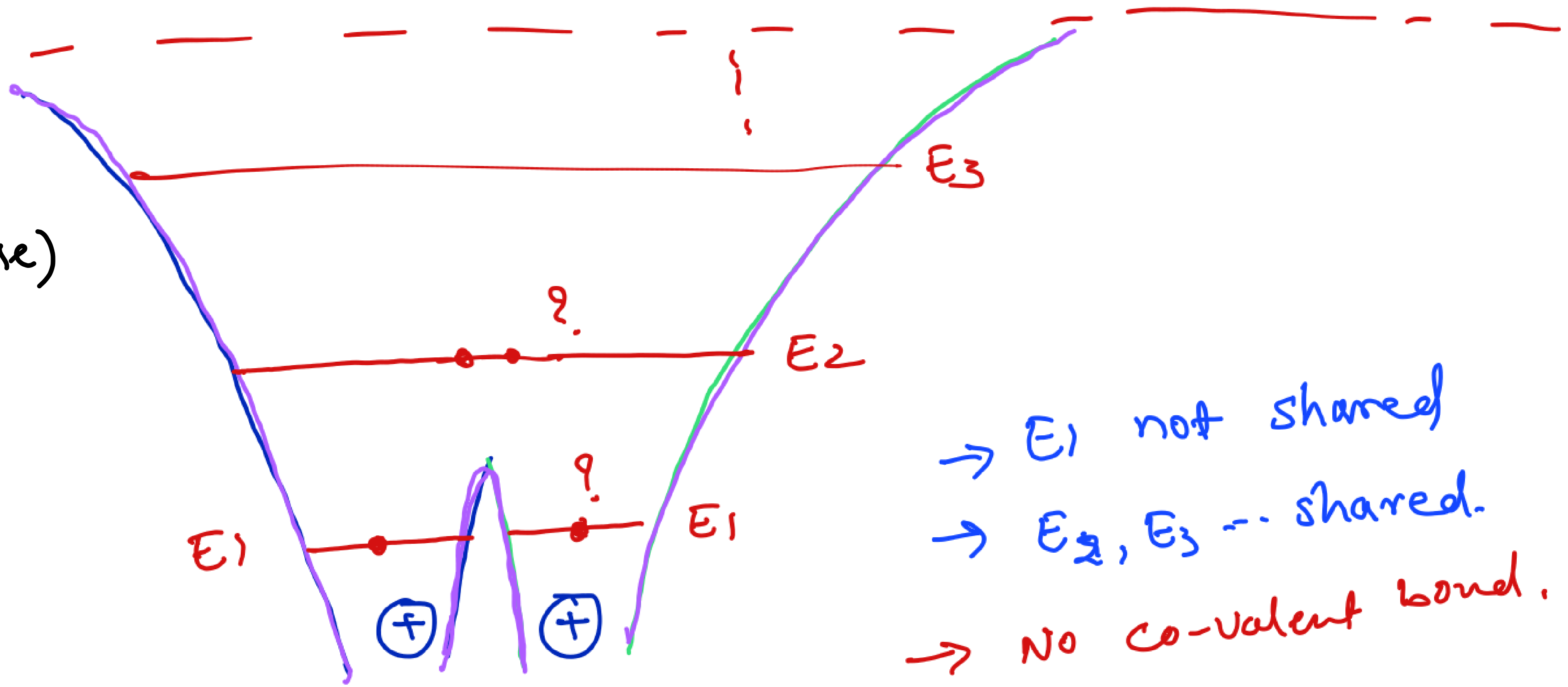


→ No sharing

→ No covalent bond.

Case 2:

Interacting atoms (close)

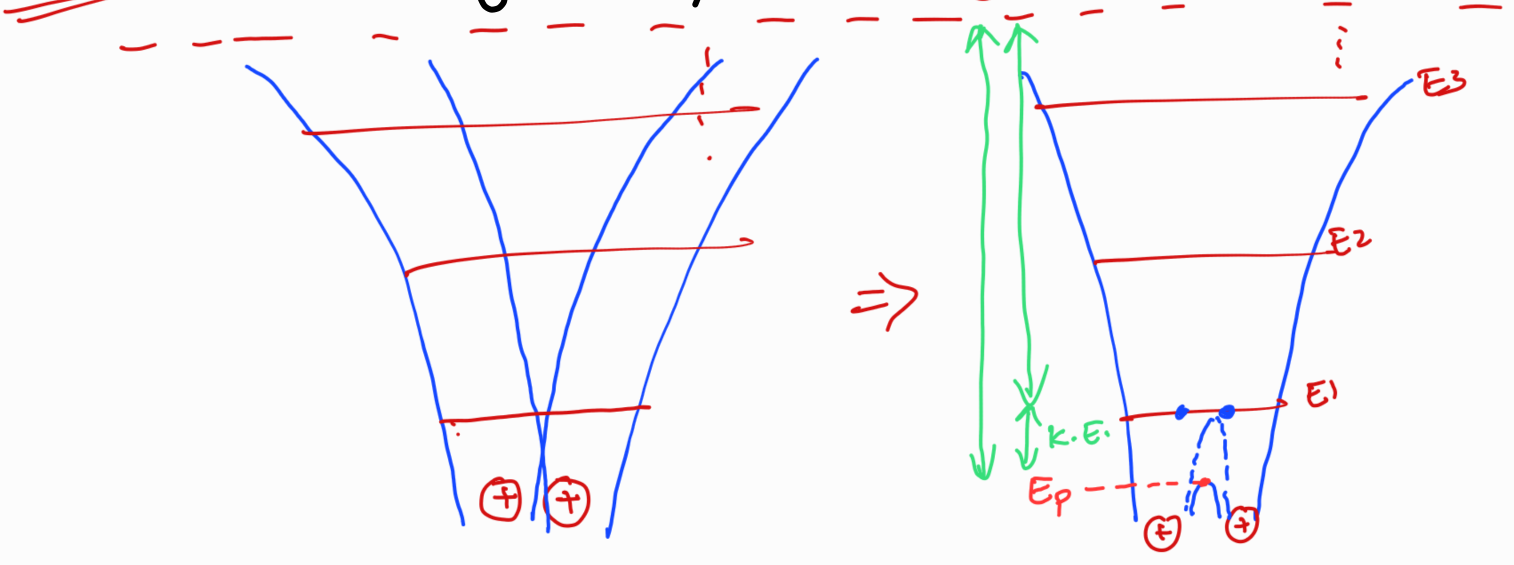


→ E_1 not shared

→ E_2, E_3 ... shared.

→ No co-valent bond.

case3 : Interacting (very close)



i) a very very small.

→ Not stable → No covalent bond.

optimum 'a' { ii) a ↑ : K.E. ↓ ⟶ 0 : covalent Bonding

" Silicon"